



Multiple endocrine diseases in cats: 15 cases (1997–2008)

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The objective of this retrospective study was to characterize a population of cats from a tertiary care center diagnosed with multiple endocrine disorders, including the specific disorders and time intervals between diagnosis of each disorder. Medical records of 15 cats diagnosed with more than one endocrine disorder were reviewed. The majority of cats were domestic shorthairs, and the mean age at the time of diagnosis of the first disorder was 10.3 years. The most common combination of disorders was diabetes mellitus and hyperthyroidism. Two cats had concurrent diabetes mellitus and hyperadrenocorticism, one cat had concurrent central diabetes insipidus and diabetes mellitus. A mean of 25.7 months elapsed between diagnoses of the first and second endocrine disorder, but this was variable. This study suggests the occurrence of multiple endocrine disorders is uncommon in cats.

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Polyendocrine glandular failure is a recognized syndrome in humans and is typically the result of immune-mediated endocrine failure.¹ The occurrence of multiple endocrinopathies in a single patient is an uncommonly recognized phenomenon in veterinary medicine. Single reports and case series document the occurrence of multiple endocrine disorders in dogs, and some such as concurrent hypothyroidism and hypoadrenocorticism may arise from a common immune-mediated etiology.^{2–9}

The occurrence of multiple endocrinopathies in cats is poorly described. In reviews of common feline endocrine conditions such as hyperthyroidism and diabetes mellitus, there is mention that these conditions can occur in the same patient.^{10–14}

This retrospective study describes 15 cats diagnosed with multiple endocrine disorders and evaluated at a referral hospital over a 12-year period. The purpose of this study was to describe the population of cats that present to a tertiary referral hospital with more than one endocrine disorder, specifically the most common combinations of endocrine disorders that occur and the timing of diagnosis of each disorder.

Materials and methods

Case selection

A computer search of the veterinary medical record database of all cats examined at the Ontario Veterinary College Teaching Hospital of the University of Guelph between January 1997 and December 2008 was performed. The database was first searched for any cats diagnosed with the following endocrine disorders: hypothyroidism, hyperthyroidism, diabetes mellitus, central diabetes insipidus, hyperadrenocorticism, hypoadrenocorticism, hypoparathyroidism, and primary hyperparathyroidism. Records of cats with a diagnosis of more than one of the previously listed disorders were then reviewed for inclusion in the study. Cases were eligible for inclusion in the study if the cat displayed compatible clinical signs, physical examination and laboratory findings, and endocrine testing, and if complete medical records were available.

Review of medical records

Data obtained from the medical records of cases included in the study consisted of signalment; weight at diagnosis; clinical signs and duration; physical examination findings; concurrent medical conditions or medications; and results of clinicopathologic testing

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including any of the following: complete blood count (CBC), serum biochemical profile, coagulation testing, urinalysis and culture, endocrine testing; diagnostic imaging; histologic and cytologic evaluations; treatments and survival time if available. When needed, referring veterinarians and owners were contacted to provide additional follow-up data.

Endocrine testing

Diabetes mellitus was diagnosed on the basis of compatible clinical signs (eg, polyuria and polydipsia), persistent fasting hyperglycemia (>11 mmol/l or 200 mg/dl), glucosuria, as well as an elevated fructosamine concentration (>340 μ mol/l) when available.¹⁰

Hyperthyroidism was diagnosed on the basis of compatible clinical signs (eg, weight loss despite polyphagia), elevated basal total thyroxine level (TT_4 , >55 nmol/l or 4.3 μ g/dl), appropriate clinical response to treatment for hyperthyroidism, and presence of a thyroid nodule when documented.¹¹

A diagnosis of pituitary-dependent hyperadrenocorticism was based on consistent clinical signs (polyuria, polydipsia, polyphagia, and fragile skin) and compatible adrenal function testing results from an adrenocorticotrophic hormone (ACTH) stimulation test and/or a low-dose dexamethasone suppression test (LDDST), as well as an elevated endogenous ACTH level (>25 pmol/l) when available. An exaggerated post-ACTH cortisol concentration (>600 nmol/l or 22 μ g/dl) 2 h after administration of synthetic cosyntropin (1.1 IU/kg [0.5 IU/lb], IV) was considered consistent with hyperadrenocorticism. Results from the LDDST were considered consistent with hyperadrenocorticism if there was inadequate suppression of cortisol concentration (>40 nmol/l or 1.4 μ g/dl) 8 h after administration of a low-dose of dexamethasone (0.1 mg/kg [0.045 mg/lb], IV).¹⁵ Presence of a pituitary adenoma along with appropriate clinical features and endocrine testing on post-mortem examination was also used to support a diagnosis of hyperadrenocorticism in one case.

Central diabetes insipidus was diagnosed based on persistent hyposthenuria (urine specific gravity <1.007), ruling out causes of secondary nephrogenic diabetes insipidus, and an appropriate response to a modified water deprivation test with desmopressin acetate therapeutic trial.¹⁶ A hypothalamic tumor identified on magnetic resonance imaging was used as supportive evidence for this diagnosis in one case.

Primary hyperparathyroidism was diagnosed based on persistent total hypercalcemia (serum calcium concentrations >2.78 mmol/l or 11.12 mg/dl) and ionized hypercalcemia (>1.34 mmol/l), low or low-normal phosphorous (reference interval (RI): 0.90–2.29 mmol/l), concurrent high or high-normal intact parathyroid hormone (PTH) concentrations (RI: 3–17 pmol/l), consistent histopathologic changes in the parathyroid mass, and no evidence of

concurrent disease that could result in hypercalcemia.¹⁷ Measurement of PTH-related protein was not performed in any of the cats included in the study.

Laboratory data

All CBC, serum biochemical, urinalysis, urine culture, and endocrinology testing was performed by the Animal Health Laboratory, University of Guelph with the exception of the intact PTH and the endogenous ACTH assays which were performed by the Diagnostic Center for Population and Animal Health, Michigan State University.

Data analysis

Data is reported as mean $\pm$ standard deviation (SD), median, and range, or frequency of occurrences and percentages. Prevalence was calculated by dividing the total number of cats with more than one endocrine disease seen at the Ontario Veterinary College Teaching Hospital over the search period (defined above) by the total number of individual cats seen at that institution during those years.

Results

Study population

During the study period, 21 cats with a diagnosis of more than one endocrinopathy were identified for potential inclusion in the study. The endocrine disorders within this population included combinations of diabetes mellitus, hyperthyroidism, central diabetes insipidus, pituitary-dependent hyperadrenocorticism, and primary hyperparathyroidism. Three cats were excluded because one endocrine disorder was iatrogenic. While having clinical signs and clinicopathologic signs of a second concurrent endocrine disorder, two cats did not meet the endocrine diagnostic inclusion criteria and were, therefore, excluded. One cat was erroneously coded as having a second endocrinopathy and was excluded from the study. The remaining 15 cases were included in the study (Table 1).

Of the 15 cats included in the study, nine (60%) were castrated males, and six (40%) were spayed females. There were no sexually intact cats included in the study. Eleven (73.3%) of the cats were domestic shorthairs (DSHs). The remaining four cats were comprised of a domestic longhair (DLH), Maine Coon, Persian, and a Turkish Angora cross.

A total of 9702 individual cats were admitted to the hospital during the study period, and the cats in this study comprised 0.15% of the feline patient admissions. Of the cats admitted to the hospital, 365 individual cats were considered to have an endocrine disorder. The prevalence of multiple endocrine disorders in the total population of feline endocrine patients was 4.1%. In the population of cats admitted to the hospital during the study period, 66% were

Table 1. Summary of cats with multiple endocrine disorders seen at the Ontario Veterinary College Teaching Hospital.

Breed	Sex	Age at first diagnosis	First endocrinopathy	Second endocrinopathy	Time to second endocrinopathy (months)
DSH	MC	6.9	Diabetes mellitus	Hyperthyroidism	1.3
DSH	FS	13.3	Diabetes mellitus	Hyperthyroidism	20.5
DSH	MC	13.9	Diabetes mellitus	Hyperthyroidism	25.2
DSH	FS	12.9	Diabetes mellitus	Hyperthyroidism	26
DSH	MC	7	Diabetes mellitus	Hyperthyroidism	77
DLH	FS	12	Hyperthyroidism	Diabetes mellitus	10
Angora-cross	MC	12.1	Hyperthyroidism	Diabetes mellitus	10
DSH	MC	10.1	Hyperthyroidism	Diabetes mellitus	18
DSH	MC	14.2	Hyperthyroidism	Diabetes mellitus	22
DSH	MC	7.8	Hyperthyroidism	Diabetes mellitus	38
Maine Coon	MC	10.8	Hyperthyroidism	Diabetes mellitus	42
Persian	FS	12.5	Diabetes mellitus	Hyperadrenocorticism	2
DSH	FS	7	Diabetes mellitus	Hyperadrenocorticism	12
DSH	MC	8	Central diabetes insipidus	Diabetes mellitus	3
DSH	FS	14.7	Hyperthyroidism	Hyperparathyroidism	17.4

MC = castrated male; FS = spayed female.

DSHs. Gender distribution of the cats admitted to the hospital during the study period was 54% male and 46% female.

The mean age of cats in this study at the time of the first endocrine disorder was 10.9 ± 2.9 years (median, 12 years; range, 7–14.2 years). The mean time from diagnosis of the first endocrine disorder to the second was 21.6 ± 19.5 months (median, 18 months; range, 3–77 months).

In the study population, the most common combination of multiple endocrine disorders was hyperthyroidism and diabetes mellitus, accounting for 11/15 (73.3%) cases. Concurrent diabetes mellitus and hyperadrenocorticism was found in 2/15 (13.3%) cats. One of the 15 (6.7%) cats was diagnosed with concurrent diabetes mellitus and central diabetes insipidus; 1/15 (6.7%) was diagnosed with hyperthyroidism and hyperparathyroidism.

Discussion

To the knowledge of the authors, the current study is the largest report of cats with naturally occurring multiple endocrine disorders. Clinical findings and laboratory results at the time of diagnosis of each endocrine disorder (ie, diabetes mellitus, hyperadrenocorticism, hyperthyroidism, diabetes insipidus, and hyperparathyroidism) were consistent with previously reported findings.^{10,16–20} The prevalence of multiple endocrine disorders in feline patients is low, representing 0.15% of the total feline patient admissions and 4.1% of the cats admitted with endocrine disorders.

Most of the cats in this study were DSHs, which is likely reflective of the overall number of DSHs in the

hospital population rather than a true breed predisposition. However, larger numbers of cats with multiple endocrine disorders are needed to confirm this observation.

The mean age at the time of diagnosis of the first endocrine disorder was 10.9 years. Hyperthyroidism and diabetes mellitus, the two most common feline endocrinopathies, are most frequently diagnosed in middle-aged to older cats.^{10,11,21,22} Similar to the cats in this study, the mean age of onset of diabetes mellitus in cats is 10 years, while the mean age of onset of hyperthyroidism is 12–13 years.^{10,11}

Of the cats in this study, 60% were castrated males and 40% were spayed females. Because this is a small group, it is difficult to determine whether castrated males are truly more predisposed to developing multiple endocrine disorders. The slight predominance of males may be due to the number of cats with diabetes mellitus in this study, as castrated males are at higher risk for developing diabetes mellitus.¹⁰ During the study period, the gender distribution of cats admitted to the hospital showed a slight predominance of males (54% male, 46% female), which may have led to the increased proportion of males in this study. However, larger groups of cats with multiple endocrine disorders must be studied to determine if a significant gender predisposition exists.

Occurrence of multiple endocrine disorders is a well-reported phenomenon in humans, and typically arises from an immune-mediated process.¹ Similar reports in dogs exist,^{3,5,6} although the prevalence of multiple endocrine disorders in dogs is uncommon (Blois, SL, Unpublished data). While a common immune-mediated etiology for human (and possibly

some canine) multiple endocrine disorders exists, this is unlikely the case in cats with multiple endocrine disorders. The disorders reported in this study (hyperthyroidism, diabetes mellitus, central diabetes insipidus, hyperparathyroidism, and hyperadrenocorticism) are not thought to have an immune-mediated basis in the cat.

The most common concurrent endocrine disorders in this study were hyperthyroidism and diabetes mellitus, affecting 11/15 (73.3%) cats in this study. This was not surprising as these two disorders are the most common endocrinopathies known to affect middle-aged to older cats, and because hyperthyroidism is a commonly cited cause of insulin resistance in diabetic cats.^{10,11}

Previous studies have shown that concurrent diabetes mellitus and hyperthyroidism is not common. One study showed that 5.8% of diabetic cats had concurrent hyperthyroidism.¹² Reports of hyperthyroid cats have shown that 1.8–5.5% have concurrent diabetes mellitus.^{13,14} These findings correlate to the low prevalence of multiple endocrine disorders in cats identified by this study.

It is unlikely that diabetes mellitus and hyperthyroidism share a common underlying etiology. Feline hyperthyroidism is due to benign hyperplastic or adenomatous changes in more than 95% of cases. The etiology of this condition is unknown but is likely multifactorial and may be the result of factors including genetics, environment, and nutrition.^{11,22,23} Similar factors have not been reported in the pathogenesis of feline diabetes mellitus. In contrast to dogs and humans, most cases of feline diabetes mellitus are not likely to be caused by immune-mediated processes.²⁴ Instead, the etiology of feline diabetes mellitus is also likely multifactorial, and may include such factors as genetics, obesity, amyloidosis, pancreatitis, islet cell degeneration, and/or islet hypoplasia.^{10,21,25} Therefore, the concurrent diagnosis of hyperthyroidism and diabetes mellitus in 11 cats in this study may reflect the likelihood of a middle-aged to older cat developing each disease independently.

Hyperthyroidism may lead to insulin resistance and abnormalities in hepatic glucose metabolism. Hyperthyroid cats are reported to have normal resting insulin and glucose levels, but impaired glucose tolerance.²⁶ Additionally, recent evidence suggests that thyroid hormone increases futile hepatic glucose cycling in cats.²⁷ Of the cats in this study, 6/11 had hyperthyroidism diagnosed first; 5/11 had diabetes mellitus diagnosed first. Further research is needed to investigate a possible link between hyperthyroidism and diabetes mellitus.

Two out of 15 cats in this study were diagnosed with diabetes mellitus and subsequently with hyperadrenocorticism. In a previous study of 104 diabetic cats, 0.9% of these cats had concurrent hyperadrenocorticism.¹² Hyperadrenocorticism is an uncommon endocrine disorder in cats, but similar to dogs is

usually a pituitary-dependent process.¹⁵ As mentioned above, the etiology of feline diabetes mellitus is likely multifactorial. Hyperadrenocorticism causes marked insulin resistance in cats, and approximately 80% of cats diagnosed with hyperadrenocorticism develop diabetes mellitus.^{15,18,19} Both of the cats in this study were diagnosed first with diabetes mellitus, then with hyperadrenocorticism 2 and 12 months later. However, the hyperadrenocorticism in these cases may develop first but remain subclinical leading to an initial diagnosis of diabetes mellitus. Once the diabetes mellitus in these cats becomes difficult to control with routine insulin administration, and/or clinical signs suggestive of hyperadrenocorticism (eg, fragile skin) develop, further investigation including adrenal function testing is often pursued.

One cat in this study was diagnosed with hyperthyroidism, then with hyperparathyroidism 17.4 months later. One previous report of a similar case exists, where the two conditions were diagnosed simultaneously.²⁸ While hyperthyroidism is common in cats, hyperparathyroidism is rare.^{11,17,29,30} Most cases of feline hyperthyroidism and primary hyperparathyroidism arise from either hyperplastic or adenomatous tissue proliferation in the respective endocrine organs.^{11,17} In this case, histology of the removed parathyroid gland was inconclusive between hyperplasia or adenomatous changes; thyroid histology was not obtained. In humans, multiple endocrine neoplasia (MEN) type 2A syndrome is associated with genetic mutations causing thyroid carcinoma, bilateral pheochromocytomas, and parathyroid hyperplasia or adenomas.³¹ There was no evidence of bilateral pheochromocytomas in this cat, and the cat's clinical course and positive response to routine treatment for hyperthyroidism makes the presence of a thyroid carcinoma unlikely. Therefore, this case is not consistent with descriptions of MEN in humans.

One cat in this study had an initial diagnosis of central diabetes insipidus, followed 3 months later by a diagnosis of diabetes mellitus. In the 3 months prior to the diagnosis of diabetes mellitus, there was no history of diabetogenic drugs (eg, corticosteroids) or other causes of diabetes mellitus identified. It is unlikely that the central diabetes insipidus and diabetes mellitus share a common etiology in this case despite the short period of time between diagnoses, and instead the two endocrine disorders are considered coincidental.

It is important to recognize the possibility of more than one endocrine disorder occurring in a feline patient. Development of a second endocrinopathy can complicate the diagnosis and treatment of the first endocrinopathy. For example, fructosamine levels may be falsely decreased in hyperthyroid cats, potentially due to increased protein turnover caused by hyperthyroidism. As such, the serum fructosamine should not be solely relied upon for the diagnosis of diabetes mellitus in hyperthyroid cats.^{32,33}

The retrospective nature of this study may have introduced some important limitations to this report.

The number of cases with multiple endocrine disorders may have been underestimated due to improper record coding or incomplete medical records. Endocrine testing does have some limitations. Total T₄ levels can be suppressed by non-thyroidal illness, presumably due to euthyroid sick syndrome.^{11,13} It is possible that hyperthyroidism was not recognized in some cats with other endocrine diseases in the hospital population studied. Therefore, the number of cases with multiple endocrine diseases may have been underestimated.

Another limitation for this study was the fact that only cats referred to a tertiary care center were included. This represents a biased sample population, as tertiary care centers may be more likely to examine more patients with concurrent endocrinopathies than primary care facilities (ie, differential admission bias). However, it is not the intention of the authors to investigate the true incidence of multiple endocrinopathies in feline patients, but rather to describe the occurrence of multiple endocrine disorders in these patients.

The recognition of multiple endocrine disorders in feline patients is important for the successful management of these patients. Diagnosis of multiple endocrine disorders relies on a combination of historical, physical, and laboratory findings. The occurrence of multiple endocrine disorders appears to be uncommon in cats. The combination of diabetes mellitus and hyperthyroidism was the most common reported in this study, and the time interval between diagnoses was highly variable. Further research in this area may help identify risk factors for development of multiple endocrine disorders in cats.

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